

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering

ASSESSMENT TOOL 2– Concept Mapping

Course Code:	CSA0311	Course Title:	Data Structures
CO Assessed:	CO1 – Imitation (BL3, BL4)	Assessment:	Assessment Tool 2 – Concept Mapping
Weightage:	10% of CIA	Total Marks:	100
CO1 Statement:	Basics, Data, Data object, Data structure, Abstract Data Types (ADT), realization of ADT in 'C', Primitive and non-primitive data structure, Linear and non-linear data structure Arrays & Recursion, Performance analysis and Measurement: Time and space analysis of algorithms, Average, best and worst-case analysis.		
Student Name:		Reg. No.:	
Section / Batch:		Date:	

Code of Conduct

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____

Instructions

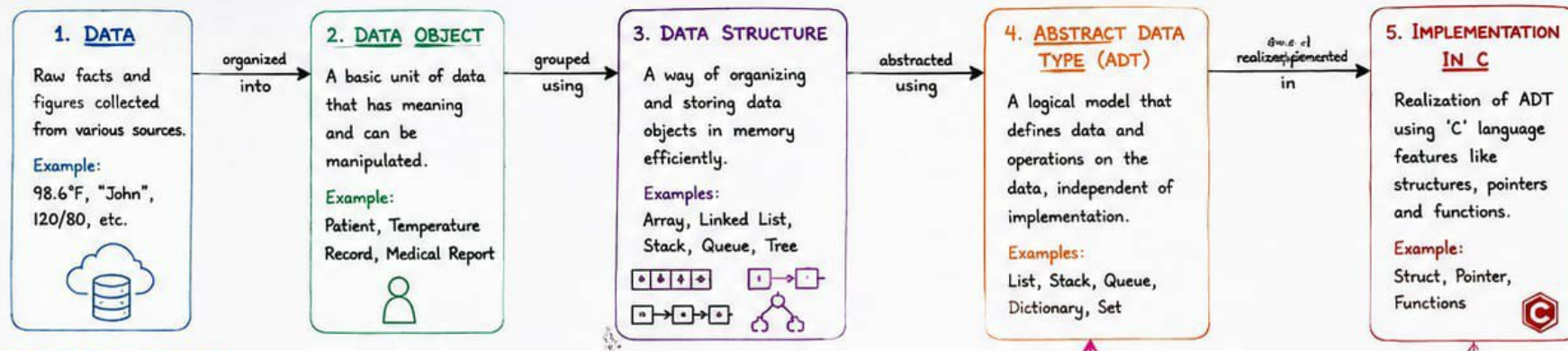
- This test contains 2 concept mapping, Total: 100 Marks.
- Evaluation of Answer Quality -Checking whether the answer includes: Concept Identification , Linkages, Organization, Integration of Literature, and Clarity
- No DTP printouts or electronic devices permitted. They should provide materials in PDF format.

Rubrics

Questions	Problem Understanding	Method Selection	Solution Process	Accuracy of Calculation	Interpretation of Results	Total Marks
Q1						
Q2						

Q1

CONCEPT MAP - Q1 : FROM DATA TO IMPLEMENTATION IN C



HOW EACH LEVEL CONTRIBUTES

- **Data** → Provides the raw information.
- **Data Object** → Gives meaning and real-world representation.
- **Data Structure** → Organizes data for efficient storage and access.
- **ADT** → Provides a clear interface and defines operations.
- **Implementation in C** → Converts the design into working code.

KEY ENABLERS IN C

STRUCTURES

Used to group different types of data into a single unit.

Example:

```
struct Patient {
    int id;
    char name[50];
    float temp;
};
```

POINTERS

Store memory addresses and enable dynamic memory allocation and efficient manipulation of data structures.

Example: `int *ptr;`
`struct Node *head;`

OPERATIONS

Functions that define the behavior of ADT.

Example:

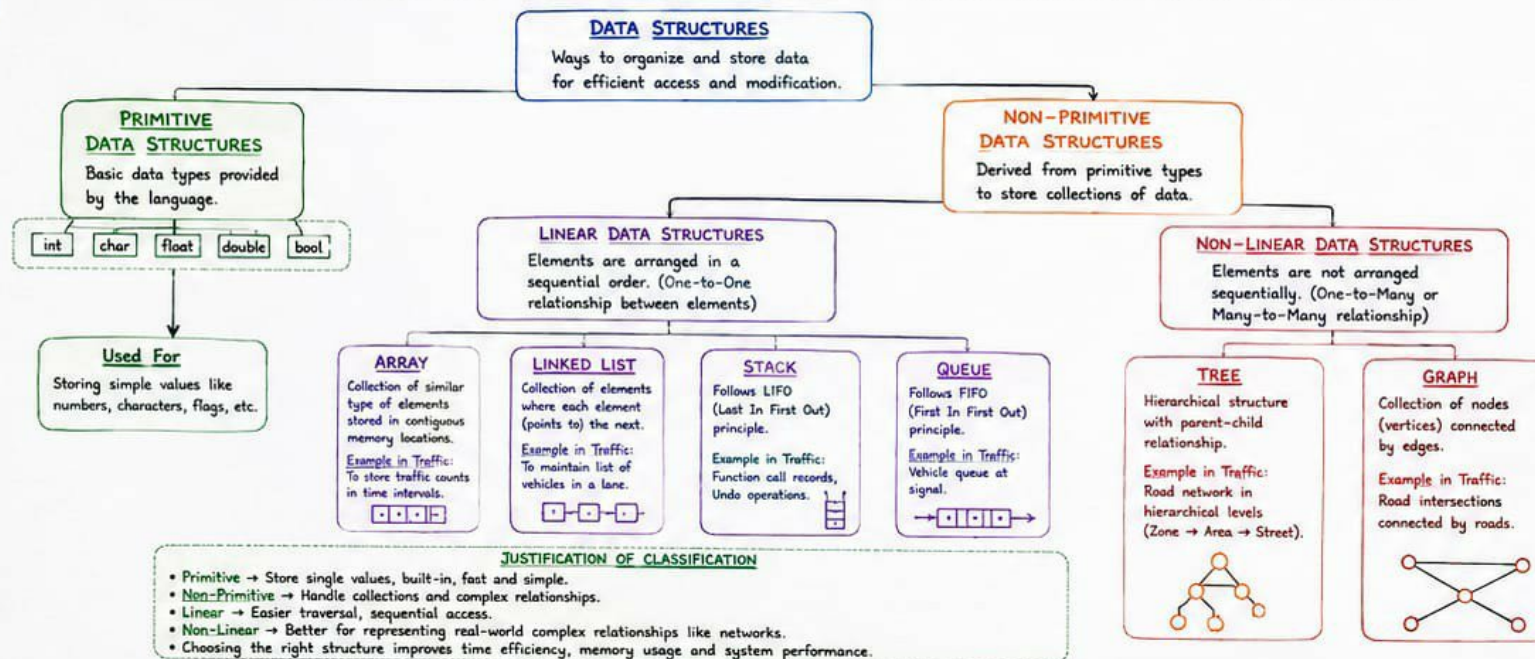
`Insert(), Delete(), Search(), Update(), Display()`

BENEFITS TO THE SYSTEM

- Efficient data organization
- Faster search and processing
- Reduced memory usage
- Code reusability and modularity
- Improved system performance and scalability

Q2

CONCEPT MAP - Q2 : FROM PRIMITIVE TO LINEAR AND NON-LINEAR STRUCTURES



APPLICATIONS IN TRAFFIC MANAGEMENT (SMART CITY)

- Arrays → Store sensor data and timing information.
- Linked Lists → Efficient insertion/deletion of vehicle records.
- Stacks → Manage traffic signal operations, function calls in software.
- Queues → Manage vehicle queues at signals.
- Trees → Represent traffic network hierarchy.
- Graphs → Find shortest path, optimize routes.